TÉCNICO
UNIVERSIDADE
DE LISBOA

0.2 · SHIP AND OCEAN SYSTEMS DESIGN

Ship Design Project

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Structure of the Project

Ten **common deliverables (D1–D10)**, extended by **two (2) design specializations** per group.

The project is group work done outside class. The in-class exercises are individual practice on a reference ship and are not graded.

COMMON TO THE WHOLE PROJECT

Baseline Project (A or B)

E.G. GROUP 1 · BASELINE A

Preliminary Stability + Damage Probabilistic Calculations

E.G. GROUP 2 · BASELINE B

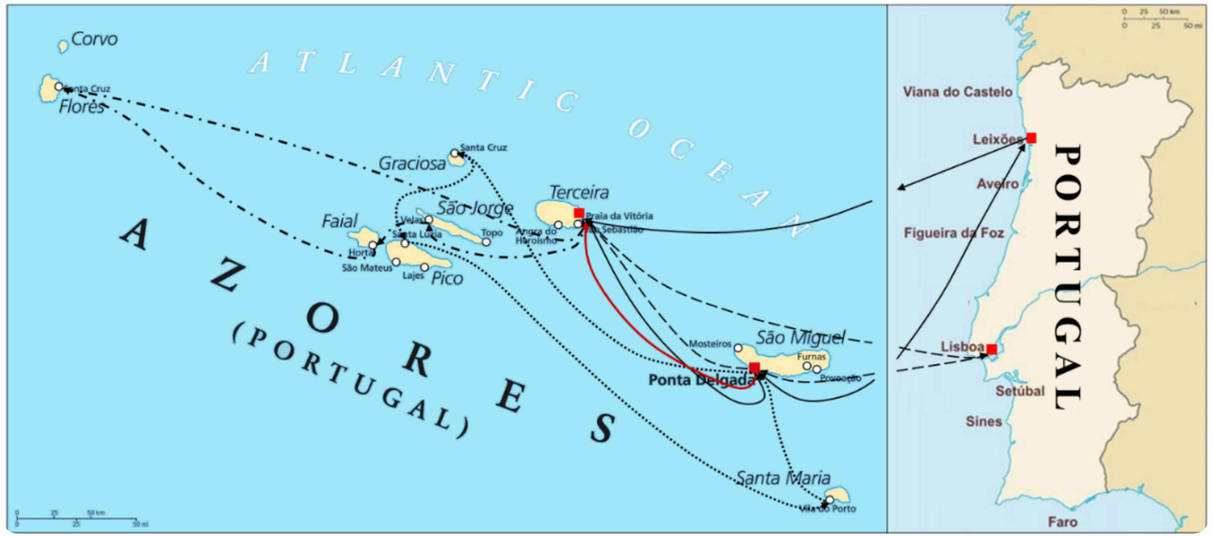
Electrical Load Balance + Alternative Fuel and Energy Storage

AVAILABLE DESIGN SPECIALIZATIONS

SP01 Longitudinal Strength Calculation	**SP02** Hull–Propeller–Engine Matching	**SP03** Electrical Load Balance	**SP04** Detailed Cost Estimation
SP05 Parametric Hull Shape Generation	**SP06** Engine Room and Steering Room Arrangement	**SP07** Preliminary Stability Calculation	**SP08** Hull Shape Optimization
SP09 Damage Probabilistic Calculations (Baseline A)	**SP10** Detailed Emissions Calculation	**SP11** Advanced Dimensional Analysis	**SP12** Alternative Fuel and Energy Storage

Two Ships for One Azores Container Service

Each group designs one of the two ships, assigned by the lecturer.





Baseline A · Azores Express

1 300 TEU · mainland to islands, weekly · dual-fuel LNG



Baseline B · Azores Feeder

130 TEU · inter-island loops · battery-electric

Azores Express: Owner’s Requirements

Ship type
Container ship, forward superstructure, aft engine room, bow thruster, good seakeeping

Cargo
Containers · gearless

Target capacity
1 300 TEU

Route
Lisbon – Ponta Delgada – Praia da Vitória – Lisbon, one round voyage a week

Port restrictions
Probably none · to be checked by each group

Service speed
18–19 kn · 7–8 kn on the generators through a PTI

Fuel and propulsion
Dual-fuel HFO/LNG, single shaft · LNG in tanks or tank containers aft

Rules and regulations
Bureau Veritas · Portuguese flag · IMO conventions and codes

Design objective
Minimum propulsion power, with one round voyage per week

Environmental targets
EEDI · CII · ECA on the route

Specialization 1
Per group

Specialization 2
Per group

Need: replaces an ageing fleet on the line. **Alternative route:** Leixões – Praia da Vitória – Ponta Delgada – Leixões, also one round voyage per week.

Azores Feeder: Owner’s Requirements

Ship type
Container ship on a large PSV-type hull (X-bow?), forward superstructure, bow thruster, good seakeeping

Cargo
Containers · geared, with its own container crane

Target capacity
130 TEU

Route
Two island loops plus one transit, all in one week (below)

Port restrictions
Probably none · to be checked by each group

Service speed
11–12 kn · 6–7 kn on the generator sets, to the mainland or in emergency

Fuel and propulsion
Swappable battery containers, aft on deck · two electric motors and shaft lines

Rules and regulations
Bureau Veritas · Portuguese flag · IMO conventions and codes

Design objective
Minimum battery capacity, with both loops in one week

Environmental targets
EEDI · CII · ECA only on trips to the mainland

Specialization 1
Per group

Specialization 2
Per group

Need: replaces an ageing fleet on the island lines. **Loop 1:** Ponta Delgada – Graciosa – Pico – Santa Maria – Ponta Delgada · **Loop 2:** Praia da Vitória – São Jorge – Faial – Flores – Praia da Vitória · plus the transit Ponta Delgada – Praia da Vitória. Batteries sized for one loop plus the transit.

Common Deliverables D1–D10

Deliver the work in a single ZIP file (Ex.: Group1_Project.zip), by e-mail, until 13 December 2026.

#	CONTENT	MUST DEMONSTRATE
D1	**Design Basis**	Owner's requirements · Port limits · Environmental targets
D2	**Reference Ship Database**	Recent similar ships · Sources · Regressions and bounds
D3	**Ship Model**	Inputs and outputs · Calculation sequence · Closed weight balance
D4	**Voyage and Financial Model**	Ship model outputs as input · Port restrictions · Revenue assumptions
D5	**Validation Note**	Reference ships · Error per quantity · Model corrected
D6	**Optimization Report**	Justified objective · Active constraints · Optimum vs requirements
D7	**Compliance Sheet**	EEDI and CII · Freeboard and stability · Draught at every port
D8	**General Arrangement and Hull Form**	A3 sketch: profile and midship section · Hydrostatics check
D9	**Specialization Documents (2)**	Calculation note · Required drawings · One per specialization
D10	**Datasheet and Presentation**	One-page A4 datasheet · Presentation of 15 minutes

Note: List of detailed deliverables will be handed out to each group on Fri 13 Nov.

Note: The class exercises are practice, not part of the project.

Specialization Content

#	DESIGN SPECIALIZATION	MUST DEMONSTRATE
SP01	**Longitudinal Strength Calculation**	Loads and bending moments · Section modulus check · Midship section drawing
SP02	**Hull–Propeller–Engine Matching**	Optimum B-series propeller · Engine matching · PTO/PTI modes · Stern drawing
SP03	**Electrical Load Balance**	Consumers per operating mode · Generator sizing · One-line diagram
SP04	**Detailed Cost Estimation**	Cost breakdown by group · Unit cost sources · Link to General Arrangement
SP05	**Parametric Hull Shape Generation**	Parametric hull definition · Coupling to the optimizer · Lines plan
SP06	**Engine Room and Steering Room Arrangement**	Machinery layout · Rudder and steering gear · Arrangement drawing
SP07	**Preliminary Stability Calculation**	Loading conditions · GZ curves against criteria · Tank and capacity plan
SP08	**Hull Shape Optimization**	Hull variables and objective · Resistance gain · Initial and optimized lines
SP09	**Damage Probabilistic Calculations**	Baseline A only · Watertight subdivision · A ≥ R · Subdivision plan
SP10	**Detailed Emissions Calculation**	Fuel use per mode · CO2, SOx, NOx per year · Machinery arrangement
SP11	**Advanced Dimensional Analysis**	Regression equations for dimensions and ratios · Fit quality and validity range
SP12	**Alternative Fuel and Energy Storage**	Energy per voyage · Tank or battery sizing · Rules check

Note: List of detailed deliverables will be handed out to each group on Fri 13 Nov.

Note: Each specialization has its own self-study deck.

Note: Your group's two are written on its Project Scope Sheet.

RESOURCES

Software

Calculations and programming

Python

High-level programming language

Google Colab or VS Code

Integrated development environment (IDE)

Excel

Spreadsheet and Solver, if Python is not used

3D modelling and drafting

Rhinoceros 3D

Surface modeling

Maxsurf Modeler

Hull form modeling

DELFTship (free)

Hull form modeling

AutoCAD

Computer Aided Design system

Note: Bring a computer to every class from Fri 6 Nov (session W1.3): each student does the exercises individually, in Python or Excel.

Groups and Questions

1

Form your group

Start now. Groups are final on **Fri 6 Nov**.

2

Ship and specializations

The lecturer assigns each group a Baseline ship (A or B) and two specializations by **13 Nov**. Keen on a particular ship type? Make your case.

3

Your questions

On the project, now. Later: office hours, one 30-minute slot per group per week, at calendly.com/vnsd-luisbatista/meetme.

Everything for the course is on the course site: l-u-i-s-b-a-t-i-s-t-a.github.io/PNSO-2026

Note: 0.3 Report Template, 0.4 Datasheet Template and 0.5 Project Scope Sheet are on the course site; your group's filled-in Scope Sheet is handed out on Fri 13 Nov.